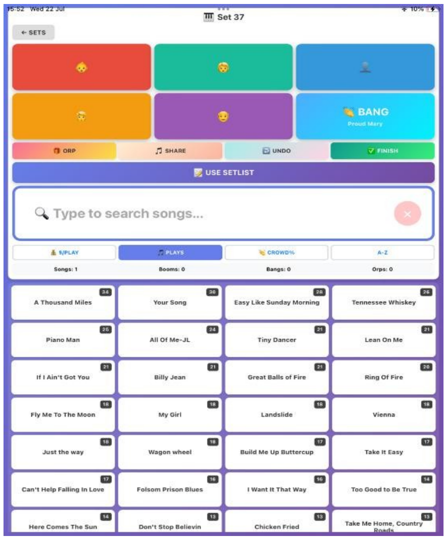
The Stowaway Piano Player Data Project

- By Samuel Martin

I've been a piano entertainer for most of my working life. The last 8 years have been mainly dedicated to playing piano and singing in piano bars on Royal Caribbean ships. The contract I'm on now is a different role, travelling piano man or 'stowaway'. The job involves playing piano all over the ship, doing pop-up performances and playing to whoever is walking by, essentially busking on a cruise ship. I'll be on the Utopia of the Seas for 13 weeks. The most profitable sets I do in terms of tips are in the elevator, yes that's right, I play sets in the elevator! People love these performances; they get very excited. These sets provide a unique opportunity for data collection and analysis. People are getting on the elevator, staying for a few minutes (sometimes more), and then they get off. I have about thirty seconds to play something that gets their attention, to get them singing, laughing or just enjoying the ride! Sometimes playing a quick chorus of something I suspect they know and like works, sometimes just playing a showpiece on the piano or improvising some jazz works, sometimes saying something silly, sometimes playing something the kids will like works, sometimes just playing whatever I feel like. Another great thing about the elevator sets is that people tip, this ship is out of Florida where I know from experience people are generous. As I'm interested in data analytics it gave me the idea to turn this job into a project.

The questions I'll be asking, does song choice matter, does crowd reaction correlate to tips earned. I'll also be using any other insights to try to increase the crowd reaction I get in the elevator and increase the number of tips I get.

The first problem to solve was the method in which I would collect data. For good quality data I need to track when a song gets a tip and when a song gets a crowd response. This was the biggest challenge. My first attempts failed. I tried note taking, I tried taking an audio recording of the script and extracting songs from that with a code word I would say out loud that meant a tip was received. Those methods proved unworkable. I ended up designing a simple app which I enlisted Claude to build. In the app I start a set, this records the start time, then a song is selected from a list and while the song is 'active' one can press a button upon receiving a tip and the tip is recorded and attached to the song, the same for crowd reaction. It's a simple system that requires me to select a song before I play it and then simply attach a crowd reaction or a tip to that song while I'm playing, just a quick tap on the screen to track the event that we're interested in. Later iterations of the app included the approximate age of the guest tipping. I was interested to do a generational analysis – what songs do age brackets tip. At the end of the set the set is closed and the time is recorded. Then there are some other values entered, total earned in tips during the set is recorded after counting up, the location of where the set is and what port we were in that day, the subjective energy level is also tracked along with footfall level. All of these were designed to gain any insights at the end of the experiment.



This is a picture of the app. Songs are selected from the list and the buttons at the top record tips or crowd reaction events attached to that song and are exported in the data.

I then ran the app for every set I played for three weeks. I'd set up in the elevator and start a set, track every song I played, how many tips it earned, whether or not I got people singing along - I landed on the rule that if more than one person sings along, the song event earns a crowd reaction in the data.

The data from the experiment is presented below. I ran all these charts initially in SQL as a script which I have published on my GitHub.

The Dataset

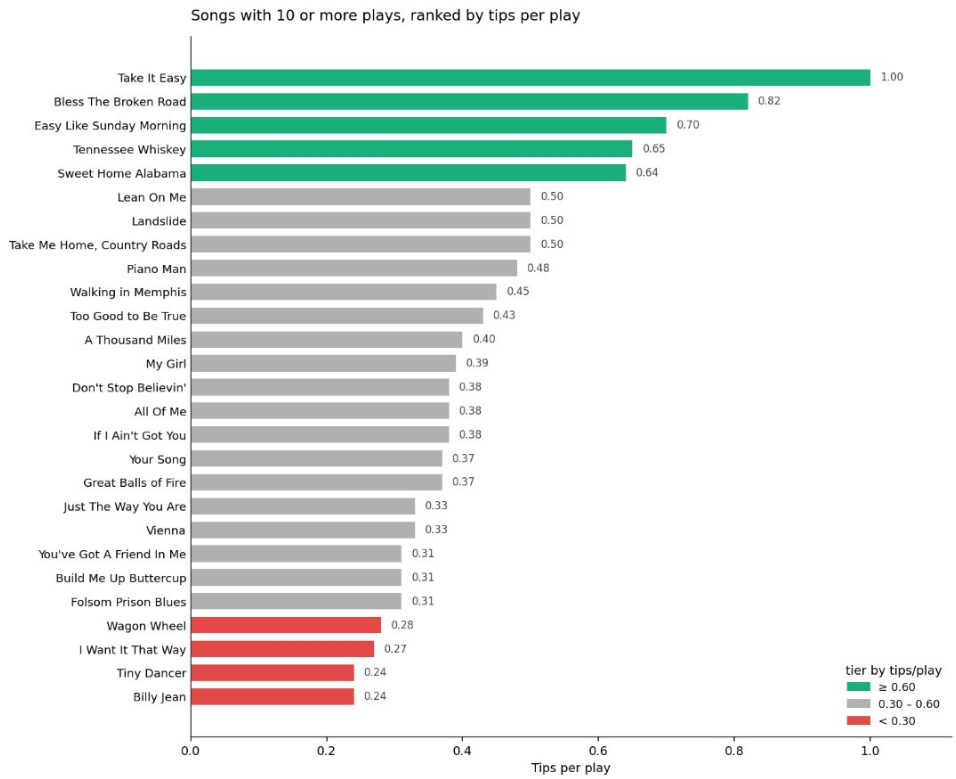
22 days · 33 sets · 42.4 hours of playing.

THE STUDY	
Sets played	33
Days played	18
Experiment length	22 days
Total hours performed	42.4
Songs played	1,047
Unique songs drawn on	202
EARNINGS	
Total cash	$3,138
Total tips	507
– Tagged to a song	375
– Orphan (song-independent)	132 (26%)
AVERAGES	
Cash per hour played	$74
Cash per set	$95
Cash per song	$3.00
Tips per set	15.4

Songs per set	31.7
Songs per hour	24.7
BASELINES	
Tips per song (all tips)	0.48
Tips per song (tagged only)	0.36

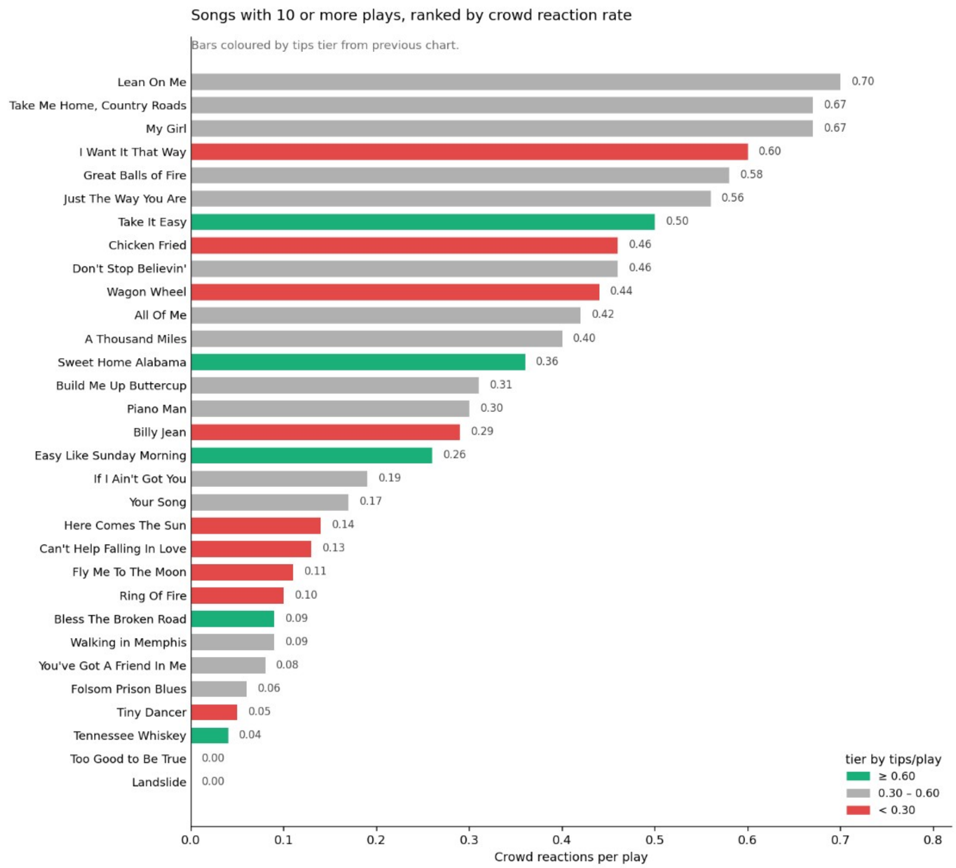
Songs ranked by tips per play

27 songs with 10 or more plays. Green tier earns 0.60+ tips per play, red under 0.30. Baseline for a random play was 0.36. Take It Easy sits at the top with 14 tips in 14 plays; against baseline, that's very unlikely to be chance (p=0.0007). The tier the song sits in is more reliable than its exact rate.



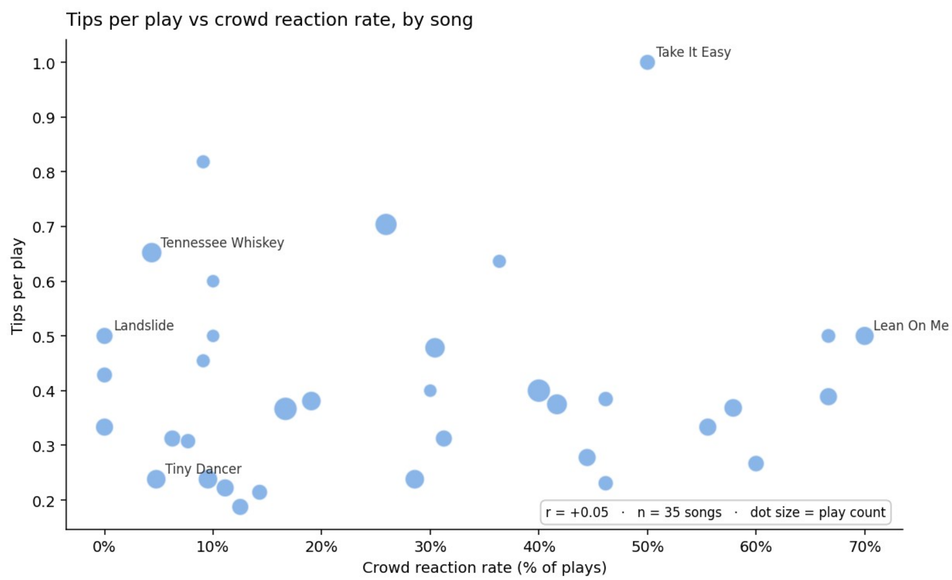
Songs ranked by crowd reaction rate

The same songs re-ranked by how often two or more people sang along. Bars are still coloured by their tips tier. Lean On Me draws the biggest response but earns mid-tier tips. Tennessee Whiskey and Bless The Broken Road earn top-tier tips but no one ever sings along.



Tips per play vs crowd reaction rate, by song

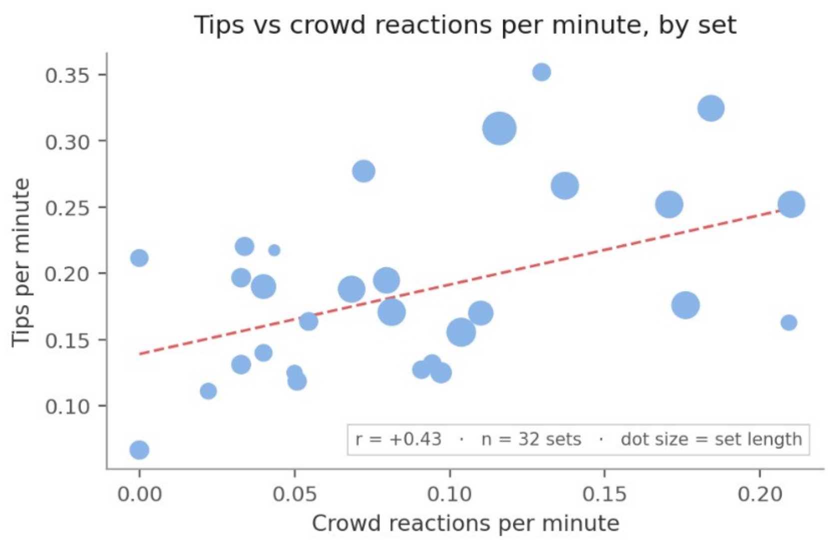
Each dot is one song, sized by play count. If crowd reaction predicted earnings the cloud would slope. It doesn't (r=0.05 across 35 songs). A play that gets a reaction does tend to earn more in the moment, 0.45 vs 0.33 average tips. That said, a song's tendency to produce reactions doesn't make it a top earner.

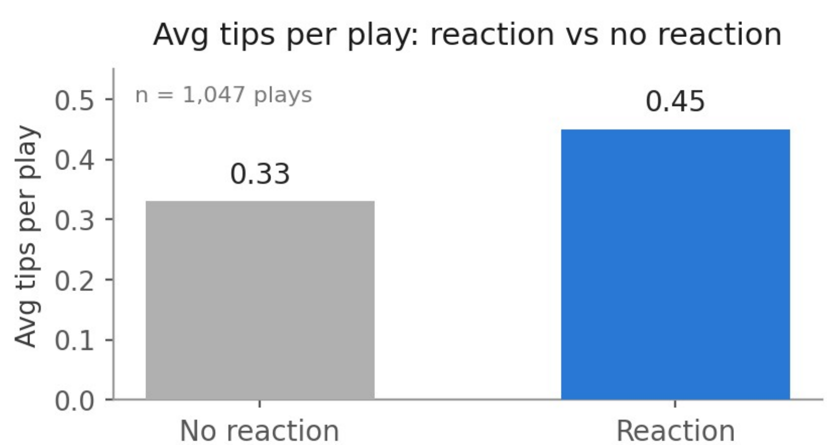


So although there are some reliably good earners and bangers, we can't depend on any one song's singalonability to reliably increase tip count. The top earners are likely a result of me playing the song to the right crowd, my intuition comes into play here and is discussed later in the analysis.

Tips vs crowd reactions per minute, by set

Each dot is one completed set, sized by set length. Unlike the song-level picture, this one does slope: sets with more crowd reactions per minute earned more tips per minute (r=0.43, p=0.013 across 32 sets). An engaged room tips more — the reaction just isn't something any particular song can reliably produce. This is correlation, not causation. A lively, generous crowd likely drives both the singing and the tipping. Or another possibility is that my ability to spot a tipping and singing group can mean I up a gear and pull out the big guns when needed, it's hard to say why but there are all sorts of reasons I could think of.



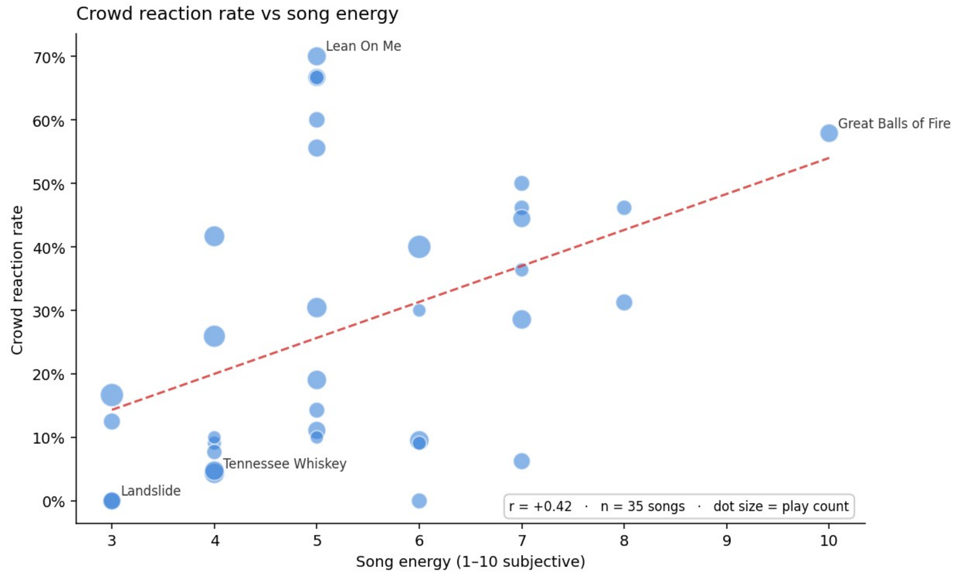


Average tips per play, with and without a crowd reaction

Across all 1,047 plays, ones with a crowd reaction averaged 0.45 tips against 0.33 without (p=0.009).

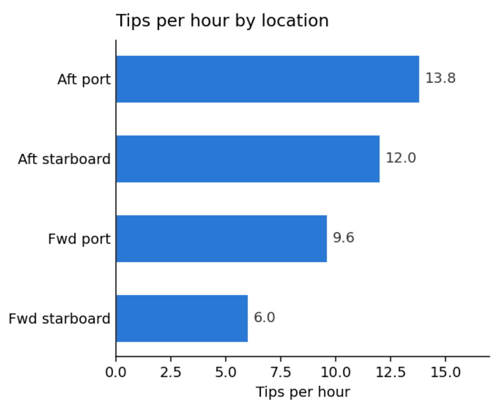
Crowd reaction rate vs song energy

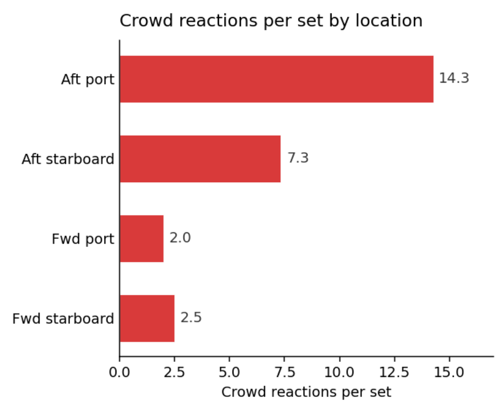
Higher-energy songs draw more crowd reactions. This trend does not carry over to tips — energy predicts singing, not tipping. I scored energy myself, so this is exploratory.



Location

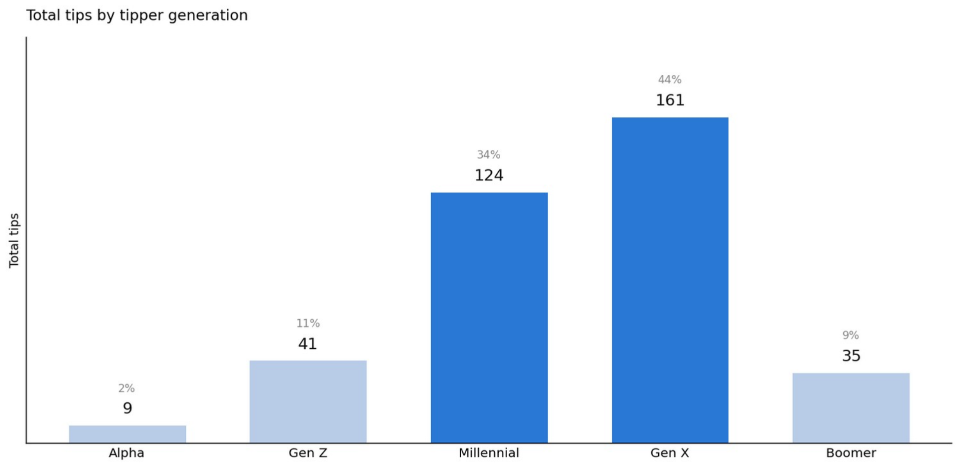
Left: tips per hour by location. Right: crowd reactions per set. Same ranking on both. Aft locations clearly outperform forward ones (12.6 vs 8.4 tips/hour, p=0.008). Port trends above starboard on both sides but the sample is too small to be sure. Fwd starboard has just 2 sets — treat as directional. That said, I knew from experience that fwd starboard performed badly which is why it was only used in the experiment a few times.





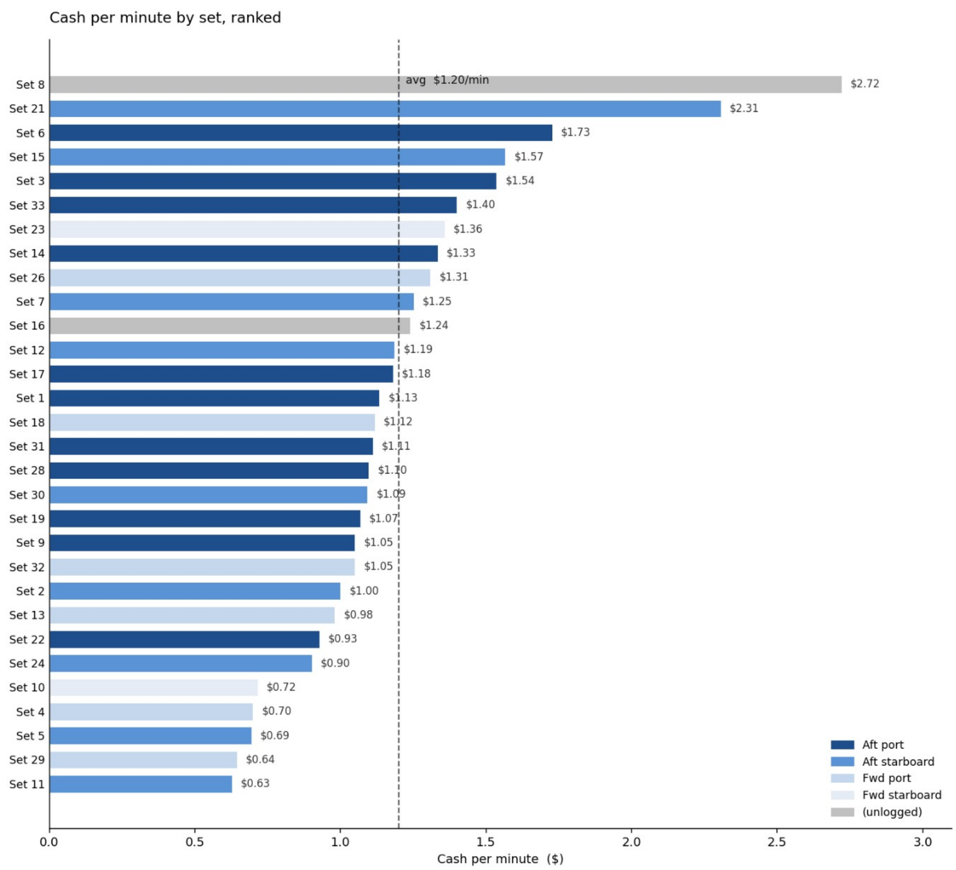
Total tips by tipper generation

Gen X and Millennials together account for 77% of tips. This is well known to me although I was surprised at how steeply the boomer contribution falls off. Access to proportions of each demographic would have been useful here, I think the graph does reflect the trends somewhat, that is to say the proportions are fairly equal so generation could predict tip count.



Cash per minute by set, ranked

All 30 completed sets, sorted by cash rate, coloured by location. Dashed line is the average across sets ($1.20/min). The range is $0.63 to $2.72 per minute.



Strategic recommendations

The clearest thing I've been able to show is that some songs are higher earners than others. I can also show that sets where the crowd sang more were more profitable — though whether the singing drives the tipping, or a good crowd simply does both, the data can't say, but my intuition tells me it's worth every bit of effort to get people singing in the moment.

It also shows that, in general, the songs I know to be crowd pleasers are. I can confidently play the high earners more and drop the ones that have performed badly. I also checked whether the ship's day port (Nassau, Cococay, Port Canaveral, or a sea day) predicted earnings. It didn't — rates were flat across all four, well within the set-to-set noise.

After years of doing this, nothing here is really new to me. There are a few songs that have performed far better than I expected and some I can confidently remove from the set. Songs that work well in a piano bar setting don't translate to top-ranked songs in the elevator. Tiny Dancer, for example, has proven unsingalongable to the cruise ship masses even though I know it to be a classic that piano bar enthusiasts love. It's these discrepancies that make this job challenging. Having such a varied range of demographics makes it hard to know what to play. As the culture of music grows it's increasingly hard to appeal to the masses with classic songs — young people don't necessarily know old songs or new hits, and older people know old songs but don't always know the massive hits we assume everyone knows. Knowing this, sticking to the piano bar seems the comfortable option, though playing to a sample more representative of the population has been fun.

I would have liked to keep running the experiment and show the recommendations paying off. The time I had was limited, and the song-level data shows the limits of song choice as a predictor of tip earnings. The problem is that my own intuition in picking a song for a specific group may be the very reason the top songs perform so well — in many cases I should just keep doing what I'm doing. If I wanted to test these songs further I'd need to play them to groups of every sort, which would be counterproductive: there's no point playing Take It Easy to a group of 13-year-olds. They won't be entertained by it, which is my job, and it won't change the fact they have no money in their pocket.

The crowd reaction findings are the most interesting, and more subtle than I expected. A song's tendency to get a reaction tells you nothing about how well it tips — the two rankings barely relate (r=0.05). But zoom out from the song and the picture changes: a play that gets a reaction earns more in that moment (0.45 vs 0.33 average tips), and a set with more reactions per minute earns more per minute (r=0.43). An engaged room tips more, it's just not something any particular song can reliably produce. Other entertainers who do this job have told me to lean on the crowd-pleasers, and playing songs that get the crowd going is exhausting — sometimes fun and natural, other times forced. These findings tell me I shouldn't worry about that so much. The engagement matters, but it can't be built from a setlist, it comes from reading the room in front of you. There are some undoubted hero songs in the piano bar repertoire, but the response and the tips come from doing your best at the time, not from following data-driven rules. If there are rules here, I'd love to know them, but on this occasion my instinct seems good enough.

Closing thoughts

So, does song choice matter, and does crowd reaction correlate to tips? Yes to both, but not quite in the way I expected going in. Song choice matters less than I assumed. My own instinct in reading the room probably does more work than the songs themselves and crowd reaction does correlate with tips. The song choice doesn't drive tips, the reaction does and often that reaction can be achieved by using certain songs I know will work. To get better data I'd need to run the experiment for far longer and try not to use my instincts at all! I'd need to play each song in the setlist a more or less equal number of times over the experiment and try to distribute each performance evenly so each one has a chance to shine under certain conditions. This would be a totally mad thing to do though. An entertainer must use his intuition and feel for the craft otherwise the whole thing becomes miserable and regimented. People already think I'm crazy for doing this, I do.

Tracking every tip and reaction mid-performance was a pain, even with the app doing most of the work. If I had a chance to do this again, knowing what I know now, I'd do several things differently with totally different aims in mind. However, I won't be doing this again, that's for sure. I think I can say for certain, no-one has done this before!

This has been a great way to bridge the wide and daunting gap I have to now face between a music career and data analytics. It has been frustrating at times but good fun. It taught me a great deal about collecting data, analysing and trying to leverage findings to deliver the optimal performance.